--- control timing results ---

cnt	avg_time	avg_send_msg_t	first_time	min	max	dev	median	correct_authorise	correct_register_value_change_%
30	0.6045262019	0.6029129028	0.7286071777	0.447511673	1.16109848	0.161739717	0.5431175232	1	1

--- stage timing averages (ms) (k,n)=(3,5) ---

stage	avg_ms
make_message	0.02863407135
total_auth_time	1.671902339
share_collect_tim	0.3776947657
key_construction	0.02058347066
message_signatu	0.17897288
zeroise_time	0.001875559489
auth_time	1.079201698
share_cost	0.5783557892
send_message	0.6558338801
total_time	2.365255356
min	1.040697098
max	27.68707275
dev	4.785330253
median	1.496553421
first_time	27.68707275

--- scenario results (k,n)=(3,5)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	2.365255356	1	1
replay_message	30	0.8390903473	1	1
wrong_target	30	0.6147782008	1	1
mismatched_valu	30	1.504516602	1	1
outdated_ts	30	0.9429057439	1	1
missing_shares	30	0.1337448756	1	1
authorisation_ski	30	0.5912939707	1	1

--- stage timing averages (ms) (k,n)=(3,9) ---

stage	avg_ms
make_message	0.03388722738
total_auth_time	2.276539803
share_collect_tim	0.5026817322
key_construction	0.02398490906
message_signatu	0.2068916957
zeroise_time	0.002272923787
auth_time	1.52156353
share_cost	0.7351239522
send_message	0.9890317917
total_time	3.313175837
min	1.282930374
max	35.94779968
dev	6.278436768
median	1.611590385
first_time	35.94779968

--- scenario results (k,n)=(3,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.313175837	1	1
replay_message	30	1.143980026	1	1
wrong_target	30	0.761961937	1	1
mismatched_valu	30	2.093362808	1	1
outdated_ts	30	1.047889392	1	1
missing_shares	30	0.2086321513	1	1
authorisation_ski	30	0.7887045542	1	1

--- stage timing averages (ms) (k,n)=(5,9) ---

stage	avg_ms
make_message	0.04719893138
total_auth_time	2.68629392
share_collect_tim	0.5657196045
key_construction	0.04436969757
message_signatu	0.3789504369
zeroise_time	0.003012021383
auth_time	1.674032211
share_cost	0.9906848272
send_message	0.9483734767
total_time	3.696656227
min	1.255750656
max	39.04676437
dev	6.781374424
median	2.414941788
first_time	39.04676437

--- scenario results (k,n)=(5,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.696656227	1	1
replay_message	30	1.022346814	1	1
wrong_target	30	1.070014636	1	1
mismatched_valu	30	2.115861575	1	1
outdated_ts	30	0.9788274765	1	1

threshold (k,n)	ratio	total node count	avg total time	correct_authorise	register_value_c	avg_scen_revok	min total time	max total time	median	total time deviatic	first time cost		increasing k	median time
control	0	0	0.6045262019	14.29%	14.29%	0	0.447511673	1.16109848	1.16109848	0.161739717	0.7286071777		(3,30)	1.435399055
(3,5)	0.6	8	2.365255356	100.00%	100.00%	0.7710549566	1.040697098	27.68707275	1.496553421	4.785330253	27.68707275		(5,30)	1.52027607
(3,9)	0.3333333333	12	3.313175837	100.00%	100.00%	1.007421811	1.282930374	35.94779968	1.611590385	6.278436768	35.94779968		(10,30)	2.081871033
(5,9)	0.5555555556	14	3.696656227	100.00%	100.00%	1.005727715	1.255750656	39.04676437	2.414941788	6.781374424	39.04676437		(15,30)	1.758933067
(3,19)	0.1578947368	22	3.003342946	100.00%	100.00%	0.9097152286	1.373529434	32.26041794	1.692771912	5.581285292	32.26041794			
(5,19)	0.2631578947	24	3.166325887	100.00%	100.00%	0.9421520763	1.368284225	36.32545471	1.858830452	6.306699279	36.32545471		increasing n	
(10,19)	0.5263157895	29	4.05523777	100.00%	100.00%	1.057716211	1.471042633	48.48790169	1.666545868	8.618502452	48.48790169		(3,5)	1.496553421
(3,30)	0.1	33	2.962152163	100.00%	100.00%	0.9361174372	1.181125641	34.19709206	1.435399055	5.981779936	34.19709206		(3,9)	1.611590385
(5,30)	0.1666666667	35	2.971076965	100.00%	100.00%	1.084362136	1.167058945	36.69500351	1.52027607	6.398957272	36.69500351		(3,19)	1.692771912
(10,30)	0.3333333333	40	4.718780518	100.00%	100.00%	1.104178694	1.594543457	57.37471581	2.081871033	10.05863282	57.37471581		(3,30)	1.435399055
(12,30)	0.4	42	3.722381592	100.00%	100.00%	1.013023323	1.516103745	59.17096138	1.675009727	10.47629495	59.17096138			
(15,30)	0.5	45	4.454930623	100.00%	100.00%	1.142715083	1.573085785	68.15695763	1.758933067	12.06989683	68.15695763			

scenario	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg	correct auth rate	correct value cha
legitimate comma	2.365255356	3.313175837	3.696656227	3.003342946	3.166325887	4.05523777	2.962152163	2.971076965	4.718780518	3.722381592	4.454930623	3.493574171	100.00%	100.00%
replay attack	0.8390903473	1.143980026	1.022346814	1.013708115	1.049939791	1.215171814	1.0242939	1.274053256	1.314727465	1.115544637	1.430980364	1.131257866	100.00%	100.00%
mismatched targ	0.6147782008	0.761961937	1.070014636	0.7830301921	0.7664124171	0.911506017	0.8143424988	0.9254773458	0.9370247523	0.7716576258	0.8845011393	0.8400642511	100.00%	100.00%
mismatched valu	1.504516602	2.093362808	2.115861575	1.832056046	1.837142309	2.01672713	1.982148488	2.321449916	2.140585581	2.199141184	2.113318443	2.014210007	100.00%	100.00%
outdated ts	0.9429057439	1.047889392	0.9788274765	1.029435794	1.107223829	1.242280006	1.106007894	1.126273473	1.239593824	1.203227043	1.435303688	1.132633469	100.00%	100.00%
missing shares	0.1337448756	0.2086321513	0.06872812907	0.05483627319	0.0795284907	0.1408179601	0.0550587972	0.07695357005	0.1209338506	0.1397530238	0.1879851023	0.1151792931		
authorisation skip	0.5912939707	0.7887045542	0.7785876592	0.7452249527	0.8126656214	0.819794337	0.6348530451	0.7819652557	0.8722066879	0.6488164266	0.8042017619	0.7525740248		

stage	control	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg	
make message	0	0.02863407135	0.0338722738	0.04719893138	0.0373840332	0.03568331401	0.0368197759	0.0340461731	0.0340461731	0.04784266154	0.03801186879	0.03605683645	0.03723736965	
share collection	0	0.3776947657	0.5026817322	0.5657196045	0.4659016927	0.5534331004	0.6670792898	0.4978100459	0.4855155945	0.7994890213	0.9924093882	0.9806156158	0.6262136228	
key reconstructio	0	0.02058347066	0.02398490906	0.04436969757	0.0245253245	0.0390291214	0.1093387604	0.02358754476	0.03661314646	0.1123905182	0.142566363	0.1971801122	0.07037899711	
CMAC constructi	0	0.17897288	0.2068916957	0.3789504369	0.2050320307	0.2035856247	0.4344860713	0.2015193303	0.2513647079	0.2482414246	0.2316792806	0.2716700236	0.2556721369	
Key Zeroising	0	0.001875559489	0.002272923787	0.003012021383	0.002225240072	0.002400080363	0.002368291219	0.002121925354	0.002153714498	0.003616015116	0.002026557922	0.002217292786	0.002389665635	
Authentication M	0	1.079201698	1.52156353	1.674032211	1.338966688	1.49658521	1.727747917	1.42103831	1.366567612	2.019715309	1.60039266	2.032558123	1.570760843	
Modbus Instructi	0.6029129028	0.6558338801	0.9890317917	0.9483734767	0.901889801	0.802342097	1.049168905	0.7559855779	0.7676919301	1.448647181	0.6895065308	0.9042183558	0.9011535934	

	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)
ratio	0.6	0.3333333333	0.5555555556	0.1578947368	0.2631578947	0.5263157895	0.1	0.1666666667	0.3333333333	0.4	0.5
node count	8	12	14	22	24	29	33	35	40	42	45
replay attack	0.8390903473	1.143980026	1.022346814	1.013708115	1.049939791	1.215171814	1.0242939	1.274053256	1.314727465	1.115544637	1.430980364
mismatched targ	0.6147782008	0.761961937	1.070014636	0.7830301921	0.7664124171	0.911506017	0.8143424988	0.9254773458	0.9370247523	0.7716576258	0.8845011393
mismatched valu	1.504516602	2.093362808	2.115861575	1.832056046	1.837142309	2.01672713	1.982148488	2.321449916	2.140585581	2.199141184	2.113318443
outdated ts	0.9429057439	1.047889392	0.9788274765	1.029435794	1.107223829	1.242280006	1.106007894	1.126273473	1.239593824	1.203227043	1.435303688
missing shares	0.1337448756	0.2086321513	0.06872812907	0.05483627319	0.0795284907	0.1408179601	0.0550587972	0.07695357005	0.1209338506	0.1397530238	0.1879851023
authorisation skip	0.5912939707	0.7887045542	0.7785876592	0.7452249527	0.8126656214	0.819794337	0.6348530451	0.7819652557	0.8722066879	0.6488164266	0.8042017619

missing_shares	30	0.06872812907	1	1
authorisation_ski	30	0.7785876592	1	1
--- stage timing averages (ms) (k,n)=(3,19) ---				
stage	avg_ms			
make_message	0.0373840332			
total_auth_time	2.052807808			
share_collect_tim	0.4659016927			
key_construction	0.0245253245			
message_signatu	0.2050320307			
zeroise_time	0.002225240072			
auth_time	1.338966688			
share_cost	0.6967782974			
send_message	0.901889801			
total_time	3.003342946			
min	1.373529434			
max	32.26041794			
dev	5.581285292			
median	1.692771912			
first_time	32.26041794			
--- scenario results (k,n)=(3,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.003342946	1	1
replay_message	30	1.013708115	1	1
wrong_target	30	0.7830301921	1	1
mismatched_valu	30	1.832056046	1	1
outdated_ts	30	1.029435794	1	1
missing_shares	30	0.05483627319	1	1
authorisation_ski	30	0.7452249527	1	1
--- stage timing averages (ms) (k,n)=(5,19) ---				
stage	avg_ms			
make_message	0.03568331401			
total_auth_time	2.313168844			
share_collect_tim	0.5534331004			
key_construction	0.0390291214			
message_signatu	0.2035856247			
zeroise_time	0.002400080363			
auth_time	1.49658521			
share_cost	0.7974704107			
send_message	0.802342097			
total_time	3.166325887			
min	1.368284225			
max	36.32545471			
dev	6.306699279			
median	1.858830452			
first_time	36.32545471			
--- scenario results (k,n)=(5,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.166325887	1	1
replay_message	30	1.049939791	1	1
wrong_target	30	0.7664124171	1	1
mismatched_valu	30	1.837142309	1	1
outdated_ts	30	1.107223829	1	1
missing_shares	30	0.0795284907	1	1
authorisation_ski	30	0.8126656214	1	1
--- stage timing averages (ms) (k,n)=(10,19) ---				
stage	avg_ms			
make_message	0.0368197759			
total_auth_time	2.958067258			
share_collect_tim	0.6670792898			
key_construction	0.1093387604			
message_signatu	0.4344860713			
zeroise_time	0.002368291219			
auth_time	1.727747917			
share_cost	1.212374369			
send_message	1.049168905			
total_time	4.05523777			
min	1.471042633			
max	48.48790169			
dev	8.618502452			
median	1.666545868			
first_time	48.48790169			
--- scenario results (k,n)=(10,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	4.05523777	1	1
replay_message	30	1.215171814	1	1
wrong_target	30	0.911506017	1	1
mismatched_valu	30	2.01672713	1	1
outdated_ts	30	1.242280006	1	1
missing_shares	30	0.1408179601	1	1

authorisation_ski	30	0.819794337	1	1	
--- stage timing averages (ms) (k,n)=(3,30) ---					
stage	avg_ms				
make_message	0.0340461731				
total_auth_time	2.161765099				
share_collect_tin	0.4978100459				
key_construction	0.02358754476				
message_signatu	0.2015193303				
zeroise_time	0.002121925354				
auth_time	1.42103831				
share_cost	0.7241805394				
send_message	0.7559855779				
total_time	2.962152163				
min	1.18125641				
max	34.19709206				
dev	5.981779936				
median	1.435399055				
first_time	34.19709206				
--- scenario results (k,n)=(3,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	2.962152163	1	1	
replay_message	30	1.0242939	1	1	
wrong_target	30	0.8143424988	1	1	
mismatched_valu	30	1.982148488	1	1	
outdated_ts	30	1.106007894	1	1	
missing_shares	30	0.0550587972	1	1	
authorisation_ski	30	0.6348530451	1	1	
--- stage timing averages (ms) (k,n)=(5,30) ---					
stage	avg_ms				
make_message	0.0340461731				
total_auth_time	2.15891997				
share_collect_tin	0.4855155945				
key_construction	0.03661314646				
message_signatu	0.2513647079				
zeroise_time	0.002153714498				
auth_time	1.366567612				
share_cost	0.7750034332				
send_message	0.7676919301				
total_time	2.971076965				
min	1.167058945				
max	36.69500351				
dev	6.398957272				
median	1.52027607				
first_time	36.69500351				
--- scenario results (k,n)=(5,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	2.971076965	1	1	
replay_message	30	1.274053256	1	1	
wrong_target	30	0.9254773458	1	1	
mismatched_valu	30	2.321449916	1	1	
outdated_ts	30	1.126273473	1	1	
missing_shares	30	0.07695357005	1	1	
authorisation_ski	30	0.7819652557	1	1	
--- stage timing averages (ms) (k,n)=(10,30) ---					
stage	avg_ms				
make_message	0.04784266154				
total_auth_time	3.205887477				
share_collect_tin	0.7994890213				
key_construction	0.1123905182				
message_signatu	0.2482414246				
zeroise_time	0.003616015116				
auth_time	2.019715309				
share_cost	1.162433624				
send_message	1.448647181				
total_time	4.718780518				
min	1.594543457				
max	57.37471581				
dev	10.05863282				
median	2.081871033				
first_time	57.37471581				
--- scenario results (k,n)=(10,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	4.718780518	1	1	
replay_message	30	1.314727465	1	1	
wrong_target	30	0.9370247523	1	1	
mismatched_valu	30	2.140585581	1	1	
outdated_ts	30	1.239593824	1	1	
missing_shares	30	0.1209338506	1	1	
authorisation_ski	30	0.8722066879	1	1	

--- stage timing averages (ms) (k,n)=(12,30) ---

stage	avg_ms
make_message	0.03801186879
total_auth_time	2.984635035
share_collect_tim	0.9924093882
key_construction	0.142566363
message_signatu	0.2316792806
zeroise_time	0.002026557922
auth_time	1.60039266
share_cost	1.368014018
send_message	0.6895065308
total_time	3.722381592
min	1.516103745
max	59.17096138
dev	10.47629495
median	1.675009727
first_time	59.17096138

--- scenario results (k,n)=(12,30)---

scenario	count	avg_time_ms	correct_authorisa	register_value_change_rate_%
legitimate_comm	30	3.722381592	1	1
replay_message	30	1.115544637	1	1
wrong_target	30	0.7716576258	1	1
mismatched_valu	30	2.199141184	1	1
outdated_ts	30	1.203227043	1	1
missing_shares	30	0.1397530238	1	1
authorisation_ski	30	0.6488164266	1	1

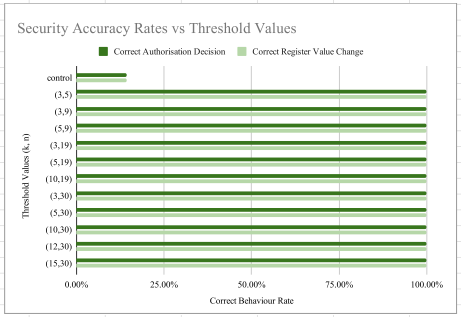
--- stage timing averages (ms) (k,n)=(15,30) ---

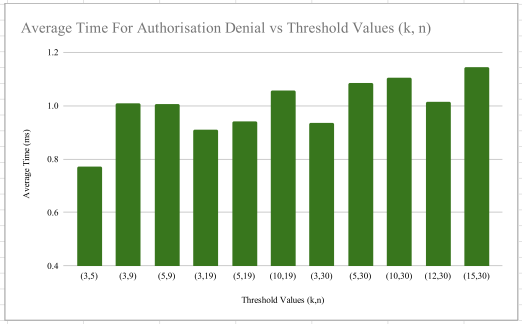
stage	avg_ms
make_message	0.03605683645
total_auth_time	3.502392769
share_collect_tim	0.9806156158
key_construction	0.1971801122
message_signatu	0.2716700236
zeroise_time	0.002217292786
auth_time	2.032558123
share_cost	1.450920105
send_message	0.9042183558
total_time	4.454930623
min	1.573085785
max	68.15695763
dev	12.06989683
median	1.758933067
first_time	68.15695763

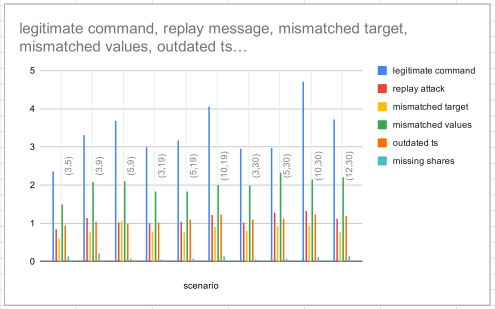
--- scenario results (k,n)=(15,30)---

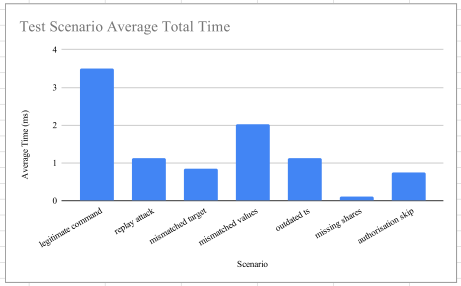
scenario	count	avg_time_ms	correct_authorisa	register_value_change_rate_%
legitimate_comm	30	4.454930623	1	1
replay_message	30	1.430980364	1	1
wrong_target	30	0.8845011393	1	1
mismatched_valu	30	2.113318443	1	1
outdated_ts	30	1.435303688	1	1
missing_shares	30	0.1879851023	1	1
authorisation_ski	30	0.8042017619	1	1

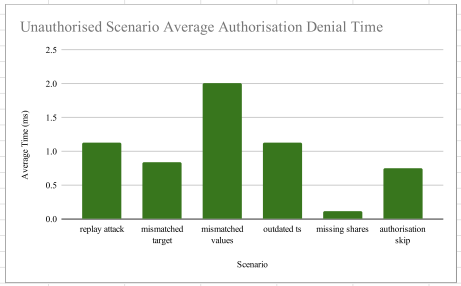
first only (ran with trial count = 1)

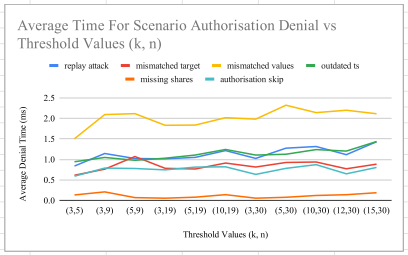

Security Accuracy Rates vs Threshold Values
Correct Authorisation Decision
Correct Register Value Change
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k, n)
0.00%
25.00%
50.00%
75.00%
100.00%
Correct Behaviour Rate

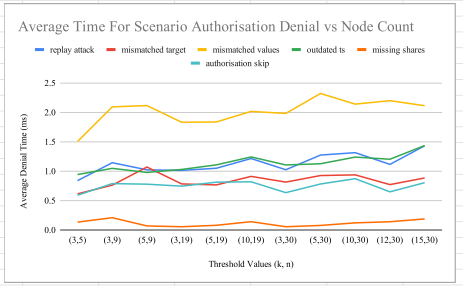

Average Time For Authorisation Denial vs Threshold Values (k, n)
Average Time (ms)
Threshold Values (k,n)

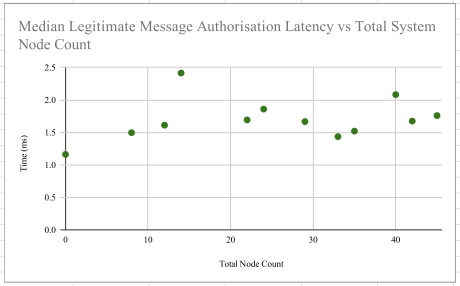

legitimate command, replay message, mismatched target, mismatched values, outdated ts…
legitimate command
replay attack
mismatched target
mismatched values
outdated ts
missing shares
scenario

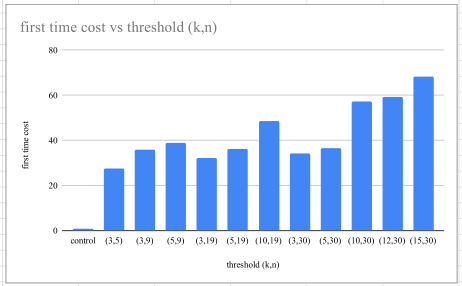

Test Scenario Average Total Time
Average Time (ms)
legitimate command
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Scenario

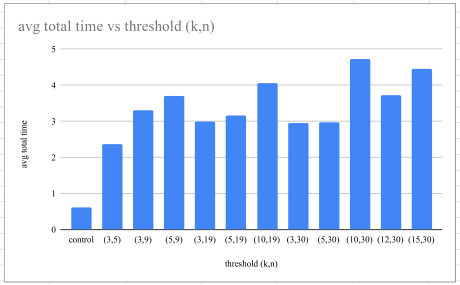

Unauthorised Scenario Average Authorisation Denial Time
Average Time (ms)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Scenario

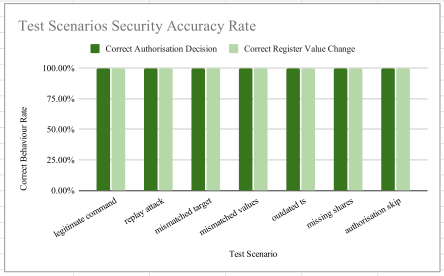

Average Time For Scenario Authorisation Denial vs Threshold Values (k, n)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Average Denial Time (ms)
Threshold Values (k, n)

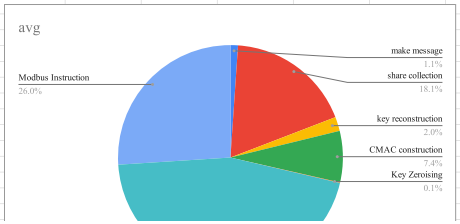

Average Time For Scenario Authorisation Denial vs Node Count
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Average Denial Time (ms)
Threshold Values (k, n)

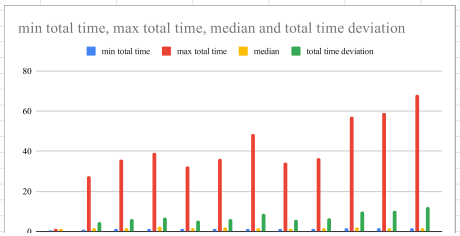

Median Legitimate Message Authorisation Latency vs Total System Node Count
Time (ms)
Total Node Count

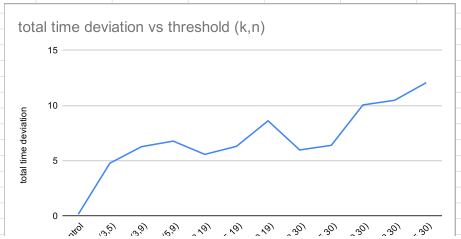

first time cost vs threshold (k,n)
first time cost
threshold (k,n)

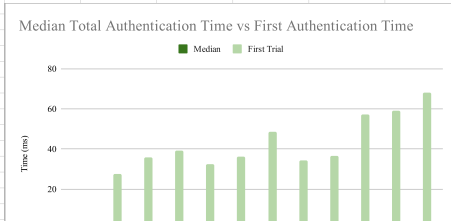

avg total time vs threshold (k,n)
avg total time
threshold (k,n)


Test Scenarios Security Accuracy Rate
Correct Authorisation Decision
Correct Register Value Change
Correct Behaviour Rate
Test Scenario

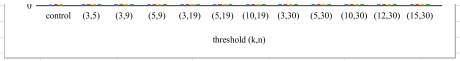

avg
make message
1.1%
share collection
18.1%
key reconstruction
2.0%
CMAC construction
7.4%
Key Zeroising
0.1%
Modbus Instruction
26.0%


min total time, max total time, median and total time deviation
min total time
max total time
median
total time deviation

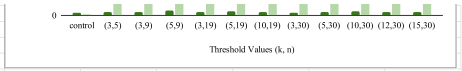

total time deviation vs threshold (k,n)
total time deviation

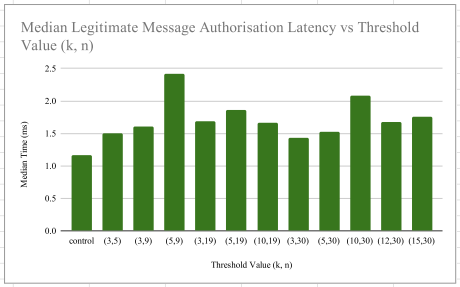

Median Total Authentication Time vs First Authentication Time
Median
First Trial
Time (ms)

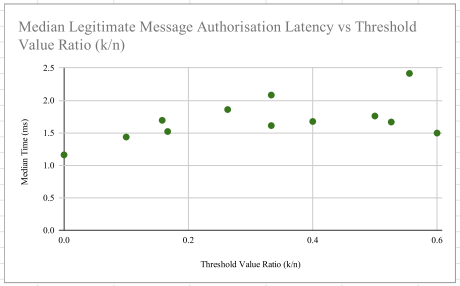

Authentication Message
45.3%

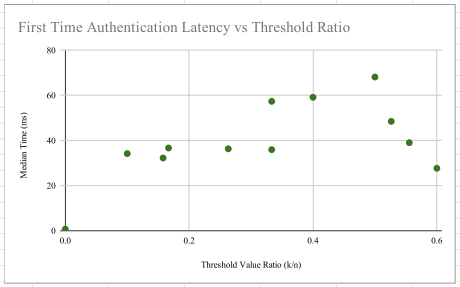

control (3,5) (3,9) (5,9) (3,19) (5,19) (10,19) (3,30) (5,30) (10,30) (12,30) (15,30)
threshold (k,n)

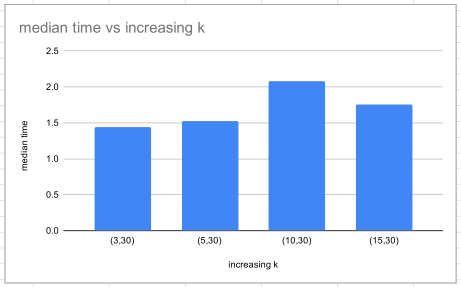

threshold (k,n)

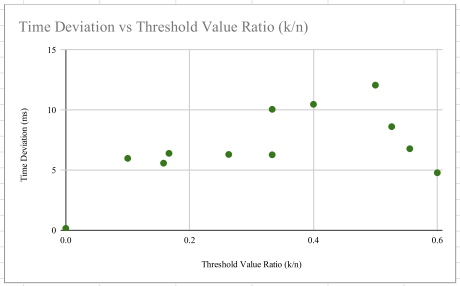

control (3,5) (3,9) (5,9) (3,19) (5,19) (10,19) (3,30) (5,30) (10,30) (12,30) (15,30)
Threshold Values (k, n)

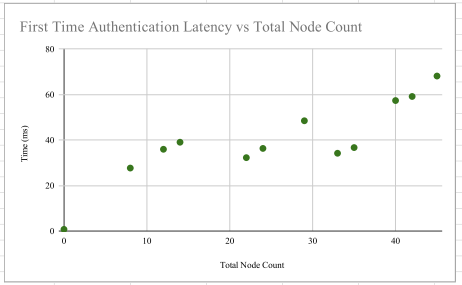

Median Legitimate Message Authorisation Latency vs Threshold Value (k, n)
Median Time (ms)
control (3,5) (3,9) (5,9) (3,19) (5,19) (10,19) (3,30) (5,30) (10,30) (12,30) (15,30)
Threshold Value (k, n)

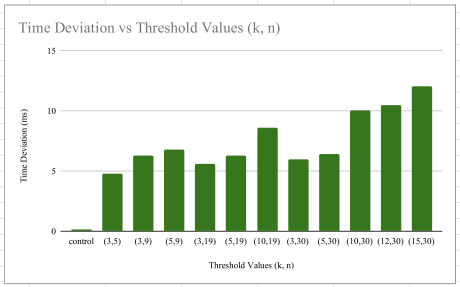

Median Legitimate Message Authorisation Latency vs Threshold Value Ratio (k/n)
Median Time (ms)
Threshold Value Ratio (k/n)

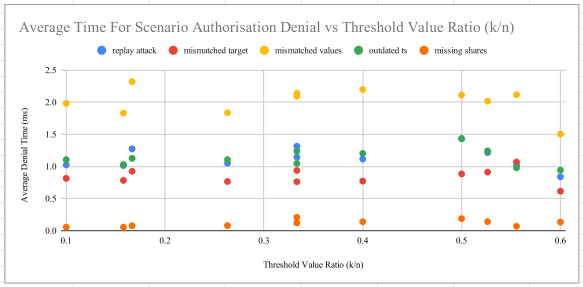

First Time Authentication Latency vs Threshold Ratio
Median Time (ms)
Threshold Value Ratio (k/n)

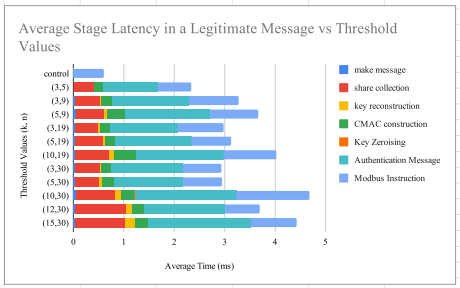

median time vs increasing k
median time
(3,30) (5,30) (10,30) (15,30)
increasing k

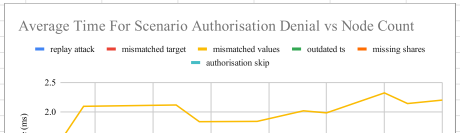

Time Deviation vs Threshold Value Ratio (k/n)
Time Deviation (ms)
Threshold Value Ratio (k/n)

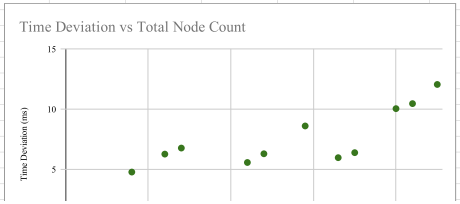

First Time Authentication Latency vs Total Node Count
Time (ms)
Total Node Count

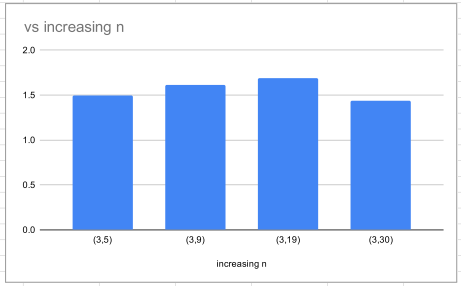

Time Deviation vs Threshold Values (k, n)
Time Deviation (ms)
control (3,5) (3,9) (5,9) (3,19) (5,19) (10,19) (3,30) (5,30) (10,30) (12,30) (15,30)
Threshold Values (k, n)

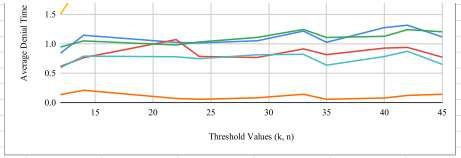

Average Time For Scenario Authorisation Denial vs Threshold Value Ratio (k/n)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
Average Denial Time (ms)
Threshold Value Ratio (k/n)

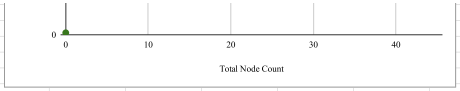

Average Stage Latency in a Legitimate Message vs Threshold Values
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k, n)
make message
share collection
key reconstruction
CMAC construction
Key Zeroising
Authentication Message
Modbus Instruction
Average Time (ms)


Average Time For Scenario Authorisation Denial vs Node Count
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip


Time Deviation vs Total Node Count
Time Deviation (ms)


vs increasing n
2.0
1.5
1.0
0.5
0.0
(3,5)
(3,9)
(3,19)
(3,30)
increasing n


Average Denial Time
1.5
1.0
0.5
0.0
15
20
25
30
35
40
45
Threshold Values (k, n)


0
0
10
20
30
40
Total Node Count

threshold (k,n)	ratio	total node count	avg total time	correct_authorise	register_value_cl	avg_scen_revok	min total time	max total time	median	total time deviatic	first time cost
control	0	0		14.29%	14.29%	0					
(3,5)	0.6	8	27.38285065	100.00%	100.00%	1.272797585	27.38285065	27.38285065	27.38285065	0	27.38285065
(3,9)	0.3333333333	12	36.04865074	100.00%	100.00%	1.095255216	36.04865074	36.04865074	36.04865074	0	36.04865074
(5,9)	0.5555555556	14	30.6994915	100.00%	100.00%	0.7225672404	30.6994915	30.6994915	30.6994915	0	30.6994915
(3,19)	0.1578947368	22	40.48871994	100.00%	100.00%	0.7706483205	40.48871994	40.48871994	40.48871994	0	40.48871994
(5,19)	0.2631578947	24	40.13061523	100.00%	100.00%	1.142978668	40.13061523	40.13061523	40.13061523	0	40.13061523
(10,19)	0.5263157895	29	45.36271095	100.00%	100.00%	0.9929736455	45.36271095	45.36271095	45.36271095	0	45.36271095
(3,30)	0.1	33	31.51321411	100.00%	100.00%	2.000212669	31.51321411	31.51321411	31.51321411	0	31.51321411
(5,30)	0.1666666667	35	35.54868698	100.00%	100.00%	1.158356667	35.54868698	35.54868698	35.54868698	0	35.54868698
(10,30)	0.3333333333	40	43.68710518	100.00%	100.00%	1.134077708	43.68710518	43.68710518	43.68710518	0	43.68710518
(12,30)	0.4	42	44.23761368	100.00%	100.00%	0.8639097214	44.23761368	44.23761368	44.23761368	0	44.23761368
(15,30)	0.5	45	62.89148331	100.00%	100.00%	0.9373823802	62.89148331	62.89148331	62.89148331	0	62.89148331

stage	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg
make message	0.02670288086	0.03623962402	0.03385543823	0.0352859497	0.0331401825	0.03385543823	0.03051757813	0.0376701355	0.03504753113	0.02908706665	0.03838539124	0.03361701965
share collection	11.51132584	15.31124115	13.07654381	15.57159424	14.75811005	17.619133	11.70229912	14.80579376	17.13776588	19.7789669	27.8980732	16.28825881
key reconstructio	0.02026557922	0.05483627319	0.02431869507	0.0319480896	0.05626678467	0.1549720764	0.02336502075	0.03814697266	0.1125335693	0.1454353333	0.1540184021	0.07419152693
CMAC constructi	0.8723735809	2.313137054	0.7436275482	2.156734467	2.002954483	1.765966415	1.091718674	1.255512238	1.070737839	1.111030579	1.128435135	1.410202547
Key Zeroising	0.002145767212	0.01263618469	0.001907348633	0.003099441528	0.009536743164	0.003814697266	0.002861022949	0.004291534424	0.003814697266	0.003099441528	0.002384185791	0.004508278587
Authentication M	14.00732994	17.46273041	15.79046249	20.05720139	21.29745483	24.10936356	17.69351959	18.33319664	24.36041832	22.08900452	32.69219398	20.71753415
Modbus Instructi	0.9064674377	0.7519721985	0.9934902191	2.568483353	1.87253952	1.628160477	0.9250640869	1.02853775	0.9229183197	1.037120819	0.9391307831	1.233989542

--- stage timing averages (ms) (k,n)=(3,5) ---

stage	avg_ms
make_message	0.02670288086
total_auth_time	26.43370628
share_collect_tim	11.51132584
key_construction	0.02026557922
message_signatu	0.8723735809
zeroise_time	0.002145767212
auth_time	14.00732994
share_cost	12.40587234
send_message	0.9064674377
total_time	27.38285065
min	27.38285065
max	27.38285065
dev	0
median	27.38285065
first_time	27.38285065

--- scenario results (k,n)=(3,5)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	1	27.38285065	1	1
replay_message	1	0.8325576782	1	1
wrong_target	1	1.803398132	1	1
mismatched_valu	1	2.385139465	1	1
outdated_ts	1	1.472473145	1	1
missing_shares	1	0.09346008301	1	1
authorisation_ski	1	1.049757004	1	1

--- stage timing averages (ms) (k,n)=(3,9) ---

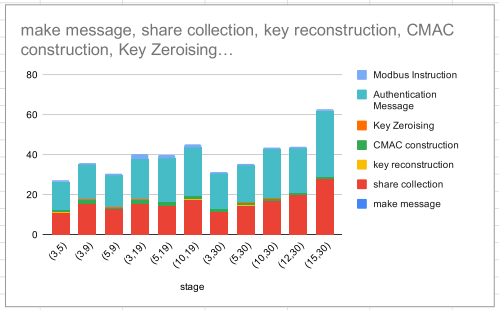
stage	avg_ms
make_message	0.03623962402
total_auth_time	35.24589539
share_collect_tim	15.31124115
key_construction	0.05483627319
message_signatu	2.313137054
zeroise_time	0.01263618469
auth_time	17.46273041
share_cost	17.68922806
send_message	0.7519721985
total_time	36.04865074
min	36.04865074
max	36.04865074
dev	0
median	36.04865074
first_time	36.04865074

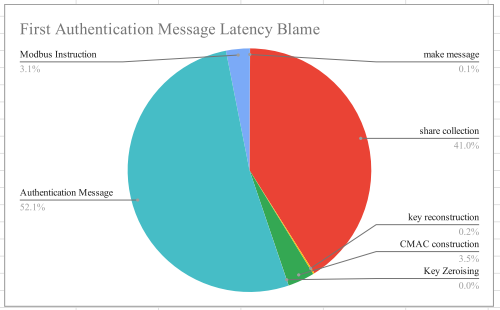
--- scenario results (k,n)=(3,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	1	36.04865074	1	1
replay_message	1	1.008272171	1	1
wrong_target	1	1.61242485	1	1
mismatched_valu	1	2.153635025	1	1
outdated_ts	1	0.9205341339	1	1
missing_shares	1	0.06699562073	1	1
authorisation_ski	1	0.8096694946	1	1

--- stage timing averages (ms) (k,n)=(5,9) ---

stage	avg_ms
make_message	0.03385543823
total_auth_time	29.65521812
share_collect_tim	13.07654381
key_construction	0.02431869507
message_signatu	0.7436275482
zeroise_time	0.001907348633
auth_time	15.79046249
share_cost	13.8463974
send_message	0.9934902191
total_time	30.6994915
min	30.6994915
max	30.6994915
dev	0
median	30.6994915
first_time	30.6994915





--- scenario results (k,n)=(5,9)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	30.6994915	1	1	
replay_message	1	0.8709430695	1	1	
wrong_target	1	0.5950927734	1	1	
mismatched_valu	1	1.428842545	1	1	
outdated_ts	1	0.8502006531	1	1	
missing_shares	1	0.06127357483	1	1	
authorisation_ski	1	0.529050827	1	1	
--- stage timing averages (ms) (k,n)=(3,19) ---					
stage	avg_ms				
make_message	0.03528594971				
total_auth_time	37.85276413				
share_collect_tim	15.57159424				
key_construction	0.0319480896				
message_signatu	2.156734467				
zeroise_time	0.003099441528				
auth_time	20.05720139				
share_cost	17.76480675				
send_message	2.568483353				
total_time	40.48871994				
min	40.48871994				
max	40.48871994				
dev	0				
median	40.48871994				
first_time	40.48871994				
--- scenario results (k,n)=(3,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	40.48871994	1	1	
replay_message	1	0.9400844574	1	1	
wrong_target	1	0.7009506226	1	1	
mismatched_valu	1	1.524925232	1	1	
outdated_ts	1	0.9422302246	1	1	
missing_shares	1	0.1020431519	1	1	
authorisation_ski	1	0.4136562347	1	1	
--- stage timing averages (ms) (k,n)=(5,19) ---					
stage	avg_ms				
make_message	0.0331401825				
total_auth_time	38.18941116				
share_collect_tim	14.75811005				
key_construction	0.05626678467				
message_signatu	2.002954483				
zeroise_time	0.009536743164				
auth_time	21.29745483				
share_cost	16.8223381				
send_message	1.87253952				
total_time	40.13061523				
min	40.13061523				
max	40.13061523				
dev	0				
median	40.13061523				
first_time	40.13061523				
--- scenario results (k,n)=(5,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	40.13061523	1	1	
replay_message	1	1.158237457	1	1	
wrong_target	1	0.8194446564	1	1	
mismatched_valu	1	1.890420914	1	1	
outdated_ts	1	2.09736824	1	1	
missing_shares	1	0.1137256622	1	1	
authorisation_ski	1	0.7786750793	1	1	
--- stage timing averages (ms) (k,n)=(10,19) ---					
stage	avg_ms				
make_message	0.03385543823				
total_auth_time	43.68495941				
share_collect_tim	17.619133				
key_construction	0.1549720764				
message_signatu	1.765966415				
zeroise_time	0.003814697266				
auth_time	24.10936356				
share_cost	19.54483986				
send_message	1.628160477				
total_time	45.36271095				

min	45.36271095				
max	45.36271095				
dev	0				
median	45.36271095				
first_time	45.36271095				
--- scenario results (k,n)=(10,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	1	45.36271095	1	1	
replay_message	1	0.9248256683	1	1	
wrong_target	1	0.7219314575	1	1	
mismatched_valu	1	1.580715179	1	1	
outdated_ts	1	0.953912735	1	1	
missing_shares	1	0.1373291016	1	1	
authorisation_ski	1	1.639127731	1	1	
--- stage timing averages (ms) (k,n)=(3,30) ---					
stage	avg_ms				
make_message	0.03051757813				
total_auth_time	30.53832054				
share_collect_tim	11.70229912				
key_construction	0.02336502075				
message_signatu	1.091718674				
zeroise_time	0.002861022949				
auth_time	17.69351959				
share_cost	12.82024384				
send_message	0.9250640869				
total_time	31.51321411				
min	31.51321411				
max	31.51321411				
dev	0				
median	31.51321411				
first_time	31.51321411				
--- scenario results (k,n)=(3,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	1	31.51321411	1	1	
replay_message	1	2.438545227	1	1	
wrong_target	1	2.114057541	1	1	
mismatched_valu	1	3.739356995	1	1	
outdated_ts	1	2.335309982	1	1	
missing_shares	1	0.1277923584	1	1	
authorisation_ski	1	1.246213913	1	1	
--- stage timing averages (ms) (k,n)=(5,30) ---					
stage	avg_ms				
make_message	0.0376701355				
total_auth_time	34.46555138				
share_collect_tim	14.80579376				
key_construction	0.03814697266				
message_signatu	1.255512238				
zeroise_time	0.004291534424				
auth_time	18.33319664				
share_cost	16.10302925				
send_message	1.02853775				
total_time	35.54868698				
min	35.54868698				
max	35.54868698				
dev	0				
median	35.54868698				
first_time	35.54868698				
--- scenario results (k,n)=(5,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	1	35.54868698	1	1	
replay_message	1	2.044439316	1	1	
wrong_target	1	0.9391307831	1	1	
mismatched_valu	1	2.043008804	1	1	
outdated_ts	1	1.239061356	1	1	
missing_shares	1	0.09107589722	1	1	
authorisation_ski	1	0.5934238434	1	1	
--- stage timing averages (ms) (k,n)=(10,30) ---					
stage	avg_ms				
make_message	0.03504753113				
total_auth_time	42.709589				
share_collect_tim	17.13776588				
key_construction	0.1125335693				

message_signatu	1.070737839																								
zeroise_time	0.003814697266																								
auth_time	24.36041832																								
share_cost	18.32413673																								
send_message	0.9229183197																								
total_time	43.68710518																								
min	43.68710518																								
max	43.68710518																								
dev	0																								
median	43.68710518																								
first_time	43.68710518																								
--- scenario results (k,n)=(10,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	43.68710518	1	1																					
replay_message	1	1.441717148	1	1																					
wrong_target	1	0.7567405701	1	1																					
mismatched_valu	1	2.424240112	1	1																					
outdated_ts	1	1.229524612	1	1																					
missing_shares	1	0.1525878906	1	1																					
authorisation_ski	1	0.7996559143	1	1																					
--- stage timing averages (ms) (k,n)=(12,30) ---																									
stage	avg_ms																								
make_message	0.02908706665																								
total_auth_time	43.15185547																								
share_collect_tim	19.7789669																								
key_construction	0.1454353333																								
message_signatu	1.111030579																								
zeroise_time	0.003099441528																								
auth_time	22.08900452																								
share_cost	21.037817																								
send_message	1.037120819																								
total_time	44.23761368																								
min	44.23761368																								
max	44.23761368																								
dev	0																								
median	44.23761368																								
first_time	44.23761368																								
--- scenario results (k,n)=(12,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	44.23761368	1	1																					
replay_message	1	1.018047333	1	1																					
wrong_target	1	0.7257461548	1	1																					
mismatched_valu	1	1.63435936	1	1																					
outdated_ts	1	0.9732246399	1	1																					
missing_shares	1	0.1537799835	1	1																					
authorisation_ski	1	0.6783008575	1	1																					
--- stage timing averages (ms) (k,n)=(15,30) ---																									
stage	avg_ms																								
make_message	0.03838539124																								
total_auth_time	61.89751625																								
share_collect_tim	27.8980732																								
key_construction	0.1540184021																								
message_signatu	1.128435135																								
zeroise_time	0.002384185791																								
auth_time	32.69219398																								
share_cost	29.18314934																								
send_message	0.9391307831																								
total_time	62.89148331																								
min	62.89148331																								
max	62.89148331																								
dev	0																								
median	62.89148331																								
first_time	62.89148331																								
--- scenario results (k,n)=(15,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	62.89148331	1	1																					
replay_message	1	1.074314117	1	1																					
wrong_target	1	0.7631778717	1	1																					
mismatched_valu	1	1.657962799	1	1																					
outdated_ts	1	1.122951508	1	1																					
missing_shares	1	0.1938343048	1	1																					
authorisation_ski	1	0.8120536804	1	1																					

--- control timing results ---

cnt	avg_time	avg_send_msg_t	first_time	min	max	dev	median	correct_authorise	correct_register_value_change_%
29	0.6002475475	0.5986608308	0.6573200226	0.447511673	1.16109848	0.1654770611	0.5421638489	1	1

--- stage timing averages (ms) (k,n)=(3,5) ---

stage	avg_ms
make_message	0.02837181091
total_auth_time	0.8111164488
share_collect_tim	0.001397626153
key_construction	0.02035601386
message_signatu	0.1466849755
zeroise_time	0.001866241981
auth_time	0.6274765935
share_cost	0.1694827244
send_message	0.6439521395
total_time	1.492089239
min	1.040697098
max	1.973390579
dev	0.1694001553
median	1.486778259

--- scenario results (k,n)=(3,5)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	1.492089239	1	1
replay_message	29	0.8358955383	1	1
wrong_target	29	0.6134345614	1	1
mismatched_valu	29	1.502579656	1	1
outdated_ts	29	0.944754173	1	1
missing_shares	29	0.1366220672	1	1
authorisation_ski	29	0.5904066152	1	1

--- stage timing averages (ms) (k,n)=(3,9) ---

stage	avg_ms
make_message	0.0340363075
total_auth_time	1.145658822
share_collect_tim	0.001652487393
key_construction	0.02302794621
message_signatu	0.1752787623
zeroise_time	0.002277308497
auth_time	0.9245461431
share_cost	0.2012499448
send_message	0.9945589921
total_time	2.18784398
min	1.282930374
max	6.743431091
dev	1.227479464
median	1.554012299

--- scenario results (k,n)=(3,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_
legitimate_comm	29	2.18784398	1	1
replay_message	29	1.12321459	1	1
wrong_target	29	0.7618789015	1	1
mismatched_valu	29	2.076790251	1	1
outdated_ts	29	1.053818341	1	1
missing_shares	29	0.2140587774	1	1
authorisation_ski	29	0.7962522836	1	1

--- stage timing averages (ms) (k,n)=(5,9) ---

stage	avg_ms
make_message	0.04732197729
total_auth_time	1.479362619
share_collect_tim	0.00235130047
key_construction	0.0448309142
message_signatu	0.3625277815
zeroise_time	0.003041892216
auth_time	1.046509578
share_cost	0.4113049343
send_message	0.9364177441
total_time	2.477686981
min	1.255750656
max	6.24871254
dev	1.230813798
median	2.189040184

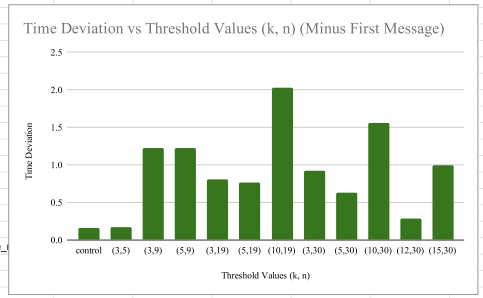
--- scenario results (k,n)=(5,9)---

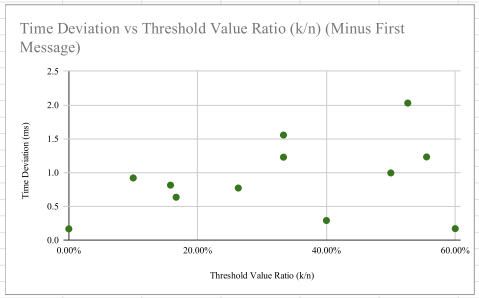
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	2.477686981	1	1
replay_message	29	1.030790395	1	1
wrong_target	29	1.079879958	1	1
mismatched_valu	29	2.112454381	1	1

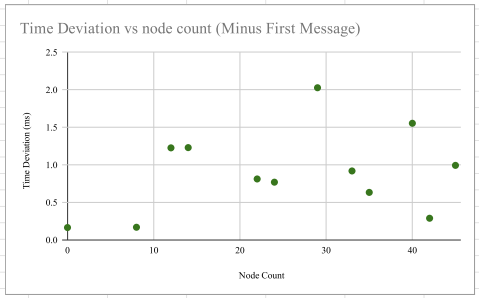
threshold (k,n)	ratio	node count	avg total time	correct_authorise	register_value_cl	avg_scen_revok	min total time	max total time	median	total time deviation
control	0.00%	0	0.6002475475	14.29%	14.29%	0	0.447511673	1.16109848	0.5421638489	0.1654770611
(3,5)	60.00%	8	1.492089239	100.00%	100.00%	0.8736831214	1.040697098	1.973390579	1.486778259	0.1694001553
(3,9)	33.33%	12	2.18784398	100.00%	100.00%	1.004335524	1.282930374	6.743431091	1.554012299	1.227479464
(5,9)	55.56%	14	2.477686981	100.00%	100.00%	1.009153224	1.255750656	6.24871254	2.189040184	1.230813798
(3,19)	15.79%	22	1.994478291	100.00%	100.00%	1.055230061	1.373529434	4.959821701	1.630783081	0.8129931529
(5,19)	26.32%	24	2.022907652	100.00%	100.00%	0.9324427309	1.368284225	4.016876221	1.830935478	0.7701756518
(10,19)	52.63%	29	2.523076945	100.00%	100.00%	1.054954255	1.471042633	9.884357452	1.64437294	2.028528348
(3,30)	10.00%	33	1.88508527	100.00%	100.00%	0.9371987705	1.181125641	4.427909851	1.432061195	0.9197514475
(5,30)	16.67%	35	1.808182947	100.00%	100.00%	1.094201515	1.167058945	3.751039505	1.502037048	0.6340737647
(10,30)	33.33%	40	2.903058611	100.00%	100.00%	1.08910292	1.594543457	6.16645813	2.067804337	1.554977555
(12,30)	0.4	42	2.258309003	100.00%	100.00%	1.01722383	1.516103745	2.625465393	1.673340797	0.2894805537
(15,30)	50.00%	45	2.258309003	100.00%	100.00%	1.140109424	1.573085785	4.966497421	1.746058464	0.9941245618

scenario	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg
legitimate_comm	1.492089239	2.18784398	2.477686981	1.994478291	2.022907652	2.523076945	1.88508527	1.808182947	2.903058611	1.810361599	2.258309003	2.123916411
replay_message	0.8358955383	1.12321459	1.030790395	0.9891740207	1.058405843	1.212744877	0.9985463373	1.288578428	1.226260744	1.120254911	1.442465289	1.120575543
wrong_target	0.6134345614	0.7618789015	1.079879958	0.7816183156	0.7720815724	0.9164070261	0.8165836334	0.9323070789	0.9450419196	0.7736189612	0.8872048608	0.8436415263
mismatched_valu	1.502579656	2.076790251	2.112454381	1.840040602	1.794132693	2.023811998	2.000471641	2.339083573	2.156685139	2.215007256	2.125041238	2.016918039
outdated_ts	0.944754173	1.053818341	0.9752799725	0.9760774415	1.068000136	1.241593525	1.115174129	1.138136305	1.243295341	1.204474219	1.384200721	1.122254937
missing_shares	0.1366220672	0.2140587774	0.06901806798	0.05509935576	0.08017441322	0.1417110706	0.05531311035	0.07718184899	0.1204835957	0.139705066	0.1883835628	0.116159176
authorisation_ski	0.5904066152	0.7962522836	0.7874965668	0.7501223992	0.8218617275	0.7934570313	0.6371037713	0.7899218592	0.8428507838	0.6502825638	0.8133608719	0.7521014976

scenario	control	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg
make message	0	0.02837181091	0.0340363075	0.04732197729	0.03741527426	0.03591899214	0.03668357586	0.03425828342	0.03406097149	0.04797146238	0.03020516757	0.03584500017	0.03655352936
share collection	0	0.001397626153	0.001652487393	0.00235130047	0.001611380742	0.001923791293	0.001619602072	0.001751143357	0.001767586017	0.00235130047	0.001446954135	0.001981340606	0.001804955701
key reconstructio	0	0.02035601386	0.02302794621	0.0448309142	0.02412138314	0.03932262289	0.1092532585	0.0235130047	0.03609986141	0.1091134959	0.1410451429	0.1926093266	0.06939027003
CMAC creation	0	0.1466849755	0.1752787623	0.3625277815	0.175147221	0.1800060272	0.3845198401	0.171439401	0.1754596316	0.1922064814	0.150557222	0.1740784481	0.2079914356
key zeroisation	0	0.001866241981	0.002273084975	0.003041892216	0.002227980515	0.002392407121	0.002252644506	0.002079996569	0.002063553909	0.003156990841	0.001940233954	0.001964897945	0.002296740732
authorisation mes	0	0.6274765935	0.9245461431	1.046509578	0.8212862344	0.9256231374	0.9100930444	0.8796083516	0.7719664738	1.148256762	0.78065642	0.9577685389	0.8903446616
Modbus instructio	0.5986608308	0.6439521395	0.9945589921	0.9364177441	0.9055877554	0.8048271311	1.051327278	0.7468174244	0.7609663338	1.36577672	0.6797642543	0.8650154903	0.8868192057



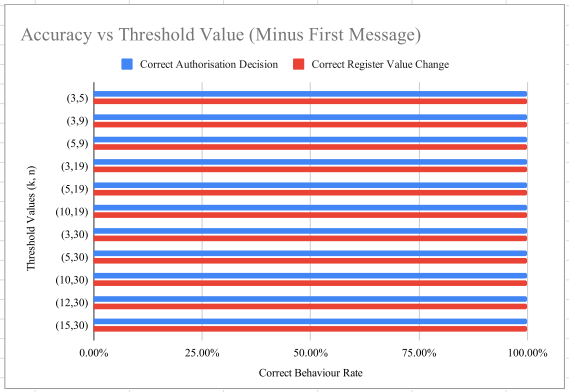


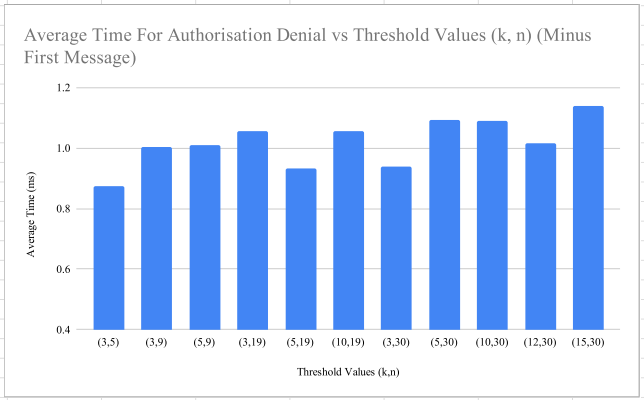


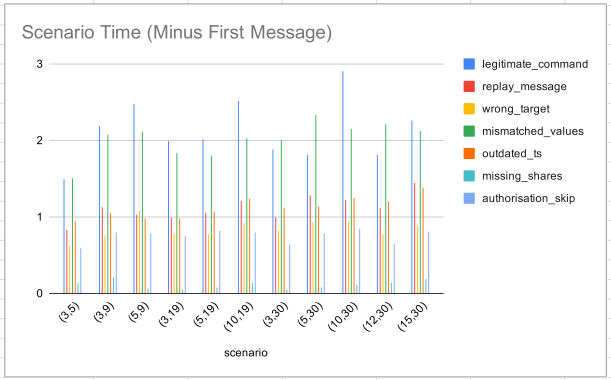
outdated_ts	29	0.9752799725	1	1
missing_shares	29	0.06901806798	1	1
authorisation_ski	29	0.7874965668	1	1
--- stage timing averages (ms) (k,n)=(3,19) ---				
stage	avg_ms			
make_message	0.03741527426			
total_auth_time	1.040360023			
share_collect_tim	0.001611380742			
key_construction	0.02412138314			
message_signatu	0.175147221			
zeroise_time	0.002227980515			
auth_time	0.8212862344			
share_cost	0.2021296271			
send_message	0.9055877554			
total_time	1.994478291			
min	1.373529434			
max	4.959821701			
dev	0.8129931529			
median	1.630783081			
--- scenario results (k,n)=(3,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	1.994478291	1	1
replay_message	29	0.9891740207	1	1
wrong_target	29	0.7816183156	1	1
mismatched_valu	29	1.840040602	1	1
outdated_ts	29	0.9760774415	1	1
missing_shares	29	0.05509935576	1	1
authorisation_ski	29	0.7501223992	1	1
--- stage timing averages (ms) (k,n)=(5,19) ---				
stage	avg_ms			
make_message	0.03591899214			
total_auth_time	1.167026059			
share_collect_tim	0.001923791293			
key_construction	0.03932262289			
message_signatu	0.1800060272			
zeroise_time	0.002392407121			
auth_time	0.9256231374			
share_cost	0.2226582889			
send_message	0.8048271311			
total_time	2.022907652			
min	1.368284225			
max	4.016876221			
dev	0.7701756518			
median	1.830935478			
--- scenario results (k,n)=(5,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	2.022907652	1	1
replay_message	29	1.058405843	1	1
wrong_target	29	0.7720815724	1	1
mismatched_valu	29	1.794132693	1	1
outdated_ts	29	1.068000136	1	1
missing_shares	29	0.08017441322	1	1
authorisation_ski	29	0.8218617275	1	1
--- stage timing averages (ms) (k,n)=(10,19) ---				
stage	avg_ms			
make_message	0.03668357586			
total_auth_time	1.424000181			
share_collect_tim	0.001619602072			
key_construction	0.1092532585			
message_signatu	0.3845198401			
zeroise_time	0.002252644506			
auth_time	0.9100930444			
share_cost	0.4967656629			
send_message	1.051327278			
total_time	2.523076945			
min	1.471042633			
max	9.884357452			
dev	2.028528348			
median	1.64437294			
--- scenario results (k,n)=(10,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	2.523076945	1	1
replay_message	29	1.212744877	1	1
wrong_target	29	0.9164070261	1	1
mismatched_valu	29	2.023811998	1	1

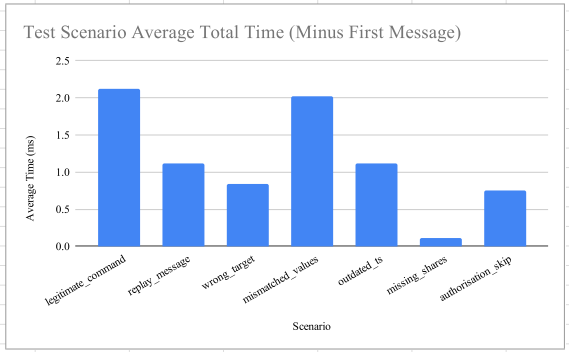
outdated_ts	29	1.241593525	1	1	
missing_shares	29	0.1417110706	1	1	
authorisation_ski	29	0.7934570313	1	1	
--- stage timing averages (ms) (k,n)=(3,30) ---					
stage	avg_ms				
make_message	0.03425828342				
total_auth_time	1.093864441				
share_collect_tim	0.001751143357				
key_construction	0.0235130047				
message_signatu	0.171439401				
zeroise_time	0.002079996569				
auth_time	0.8796083516				
share_cost	0.19794497				
send_message	0.7468174244				
total_time	1.88508527				
min	1.181125641				
max	4.427909851				
dev	0.9197514475				
median	1.432061195				
--- scenario results (k,n)=(3,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	1.88508527	1	1	
replay_message	29	0.9985463373	1	1	
wrong_target	29	0.8165836334	1	1	
mismatched_valu	29	2.000471641	1	1	
outdated_ts	29	1.115174129	1	1	
missing_shares	29	0.05531311035	1	1	
authorisation_ski	29	0.6371037713	1	1	
--- stage timing averages (ms) (k,n)=(5,30) ---					
stage	avg_ms				
make_message	0.03406097149				
total_auth_time	1.002944749				
share_collect_tim	0.001767586017				
key_construction	0.03609986141				
message_signatu	0.1754596316				
zeroise_time	0.002063553909				
auth_time	0.7719664738				
share_cost	0.2146835985				
send_message	0.7609663338				
total_time	1.808182947				
min	1.167058945				
max	3.751039505				
dev	0.6340737647				
median	1.502037048				
--- scenario results (k,n)=(5,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	1.808182947	1	1	
replay_message	29	1.288578428	1	1	
wrong_target	29	0.9323070789	1	1	
mismatched_valu	29	2.339083573	1	1	
outdated_ts	29	1.138136305	1	1	
missing_shares	29	0.07718184899	1	1	
authorisation_ski	29	0.7899218592	1	1	
--- stage timing averages (ms) (k,n)=(10,30) ---					
stage	avg_ms				
make_message	0.04797146238				
total_auth_time	1.474660018				
share_collect_tim	0.00235130047				
key_construction	0.1091134959				
message_signatu	0.1922064814				
zeroise_time	0.003156990841				
auth_time	1.148256762				
share_cost	0.3052908799				
send_message	1.36577672				
total_time	2.903058611				
min	1.594543457				
max	6.16645813				
dev	1.554977555				
median	2.067804337				
--- scenario results (k,n)=(10,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	2.903058611	1	1	
replay_message	29	1.226260744	1	1	
wrong_target	29	0.9450419196	1	1	
mismatched_valu	29	2.156685139	1	1	

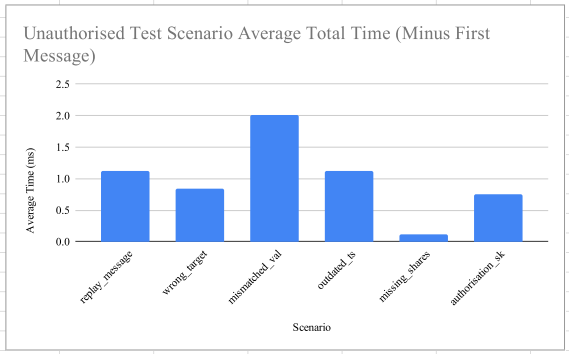
outdated_ts	29	1.243295341	1	1																						
missing_shares	29	0.1204835957	1	1																						
authorisation_ski	29	0.8428507838	1	1																						
--- stage timing averages (ms) (k,n)=(12,30) ---																										
stage	avg_ms																									
make_message	0.03020516757																									
total_auth_time	1.090378597																									
share_collect_tim	0.001446954135																									
key_construction	0.1410451429																									
message_signatu	0.150557222																									
zeroise_time	0.001940233954																									
auth_time	0.78065642																									
share_cost	0.2942660759																									
send_message	0.6797642543																									
total_time	1.810361599																									
min	1.516103745																									
max	2.625465393																									
dev	0.2894805537																									
median	1.673340797																									
--- scenario results (k,n)=(12,30)---																										
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%																					
legitimate_comm	29	1.810361599	1	1																						
replay_message	29	1.120254911	1	1																						
wrong_target	29	0.7736189612	1	1																						
mismatched_valu	29	2.215007256	1	1																						
outdated_ts	29	1.204474219	1	1																						
missing_shares	29	0.139705066	1	1																						
authorisation_ski	29	0.6502825638	1	1																						
--- stage timing averages (ms) (k,n)=(15,30) ---																										
stage	avg_ms																									
make_message	0.03584500017																									
total_auth_time	1.345519362																									
share_collect_tim	0.001981340606																									
key_construction	0.1926093266		8.53%																							
message_signatu	0.1740784481																									
zeroise_time	0.001964897945																									
auth_time	0.9577685389																									
share_cost	0.3698283228																									
send_message	0.8650154903																									
total_time	2.258309003																									
min	1.573085785																									
max	4.966497421																									
dev	0.9941245618																									
median	1.746058464																									
--- scenario results (k,n)=(15,30)---																										
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%																					
legitimate_comm	29	2.258309003	1	1																						
replay_message	29	1.442465289	1	1																						
wrong_target	29	0.8872048608	1	1																						
mismatched_valu	29	2.125041238	1	1																						
outdated_ts	29	1.384200721	1	1																						
missing_shares	29	0.1883835628	1	1																						
authorisation_ski	29	0.8133608719	1	1																						

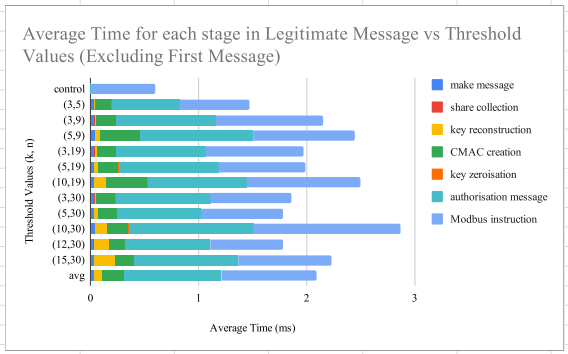

Accuracy vs Threshold Value (Minus First Message)
Correct Authorisation Decision
Correct Register Value Change
Threshold Values (k, n)
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
0.00%
25.00%
50.00%
75.00%
100.00%
Correct Behaviour Rate

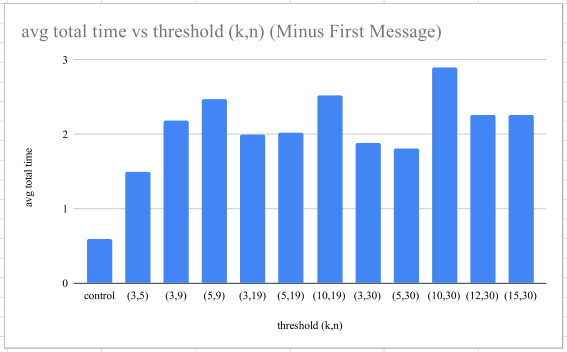

Average Time For Authorisation Denial vs Threshold Values (k, n) (Minus First Message)
Average Time (ms)
1.2
1.0
0.8
0.6
0.4
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k,n)

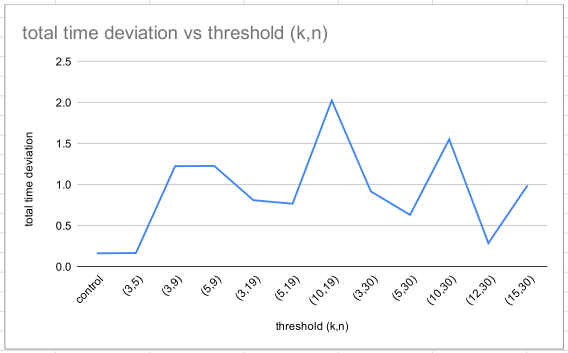

Scenario Time (Minus First Message)
legitimate_command
replay_message
wrong_target
mismatched_values
outdated_ts
missing_shares
authorisation_skip
scenario

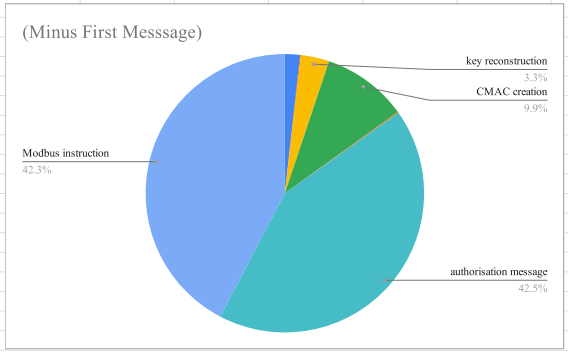

Test Scenario Average Total Time (Minus First Message)
Average Time (ms)
legitimate_command
replay_message
wrong_target
mismatched_values
outdated_ts
missing_shares
authorisation_skip
Scenario

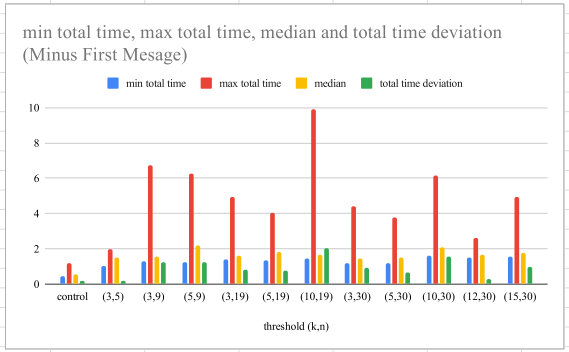

Unauthorised Test Scenario Average Total Time (Minus First Message)
Average Time (ms)
replay_message
wrong_target
mismatched_val
outdated_ts
missing_shares
authorisation_sk
Scenario


Average Time for each stage in Legitimate Message vs Threshold Values (Excluding First Message)
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
avg
Threshold Values (k, n)
make message
share collection
key reconstruction
CMAC creation
key zeroisation
authorisation message
Modbus instruction
Average Time (ms)


avg total time vs threshold (k,n) (Minus First Message)
avg total time
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
threshold (k,n)


total time deviation vs threshold (k,n)
total time deviation
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
threshold (k,n)


(Minus First Messsage)
key reconstruction
3.3%
CMAC creation
9.9%
Modbus instruction
42.3%
authorisation message
42.5%


min total time, max total time, median and total time deviation
(Minus First Mesage)
min total time
max total time
median
total time deviation
10
8
6
4
2
0
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
threshold (k,n)